

INTRODUCTION

Assignment 3 revolved around the how efficient Translation task done by three different models NLLB, IndicTrans and ChatGPT.

It is done by choosing randomly 1000 test sentences of Hindi, English and Gujarati languages for first two models and 50 random test sentences for each of the above three languages for ChatGPT model.

Answer of 2nd Question:-

-----Below are the three tables for each model's score-----

Model: NLLB

Translation Direction	Metric Type	BLEU (corpus)	BLEU (sentence)	ROUGE-1 (P/R/F1)	ROUGE-2 (P/R/F1)	ROUGE-L (P/R/F1)
English to Hindi	BLEU & ROUGE	0.6369	0.6164	0.6063 / 0.5743 / 0.5854	0.3586 / 0.3393 / 0.3458	0.5657 / 0.5357 / 0.5461
Hindi to English	BLEU & ROUGE	0.6862	0.6502	0.6237 / 0.6135 / 0.6139	0.3899 / 0.3873 / 0.3854	0.5890 / 0.5799 / 0.5801
Hindi to Gujarati	BLEU & ROUGE	0.5836	0.5768	0.5254 / 0.4848 / 0.4989	0.2774 / 0.2548 / 0.2625	0.5020 / 0.4634 / 0.4768
Gujarati to Hindi	BLEU & ROUGE	0.6387	0.6191	0.6055 / 0.5784 / 0.5869	0.3668 / 0.3526 / 0.3564	0.6055 / 0.5784 / 0.5869

Notes:

- **P/R/F1:** Precision / Recall / F1-score.
- **Metric Type:** Indicates the metric calculation method (BLEU corpus-level, BLEU sentence-level, and ROUGE).

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Model: IndicTrans

Translation Direction	Metric Type	BLEU (corpus)	BLEU (sentence)	ROUGE-1 (P/R/F1)	ROUGE-2 (P/R/F1)	ROUGE-L (P/R/F1)
English to Hindi	BLEU & ROUGE	0.6912	0.6546	0.6306 / 0.6263 / 0.6252	0.3973 / 0.3949 / 0.3939	0.5944 / 0.5900 / 0.5891
Hindi to English	BLEU & ROUGE	0.7423	0.6939	0.6597 / 0.6168 / 0.6340	0.4401 / 0.4116 / 0.4226	0.6263 / 0.5853 / 0.6018
Hindi to Gujarati	BLEU & ROUGE	0.6304	0.5962	0.5127 / 0.4676 / 0.4856	0.2599 / 0.2353 / 0.2452	0.4888 / 0.4452 / 0.4628
Gujarati to Hindi	BLEU & ROUGE	0.6630	0.6243	0.5959 / 0.5951 / 0.5913	0.3603 / 0.3615 / 0.3583	0.5622 / 0.5613 / 0.5579

Notes:

- **P/R/F1:** Precision / Recall / F1-score.
 - **Metric Type:** Indicates the metrics' method (BLEU corpus-level, BLEU sentence-level, and ROUGE).
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Model: ChatGPT

Translation Direction	Metric Type	BLEU (corpus)	BLEU (sentence)	ROUGE-1 (P/R/F1)	ROUGE-2 (P/R/F1)	ROUGE-L (P/R/F1)
English to Hindi	BLEU & ROUGE	0.7536	0.7255	0.6920 / 0.6904 / 0.6878	0.4955 / 0.4932 / 0.4918	0.6655 / 0.6630 / 0.6610
Hindi to English	BLEU & ROUGE	0.6986	0.6403	0.5982 / 0.5960 / 0.5919	0.3727 / 0.3747 / 0.3697	0.5714 / 0.5662 / 0.5640
Hindi to Gujarati	BLEU & ROUGE	0.6241	0.6046	0.5449 / 0.5210 / 0.5302	0.2719 / 0.2614 / 0.2650	0.5152 / 0.4931 / 0.5017
Gujarati to Hindi	BLEU & ROUGE	0.7282	0.7181	0.6817 / 0.6774 / 0.6767	0.4695 / 0.4665 / 0.4660	0.6456 / 0.6399 / 0.6401

Notes:

- **P/R/F1:** Precision / Recall / F1-score.
 - **Metric Type:** Indicates the metric calculation method (BLEU corpus-level, BLEU sentence-level, and ROUGE).
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Answer of 3rd Question:-

The learning from the assignment :

Before going to the analysis here are definitions of the methods used:

Corpus_BLEU :-

Corpus-BLEU is a metric for evaluating machine translations at the corpus level, using aggregated data across all sentences to compute n-gram precision.

Sentence_BLEU:-

Sentence-BLEU assesses the quality of machine translations at the individual sentence level by comparing a translated sentence to one or more reference sentences. It calculates precision for n-grams directly for each sentence and averages these across the dataset.

ROUGE_1:-

ROUGE-1 is a metric used to evaluate the quality of summaries by measuring the overlap of unigrams (single words) between the system-generated summary and a set of reference summaries.

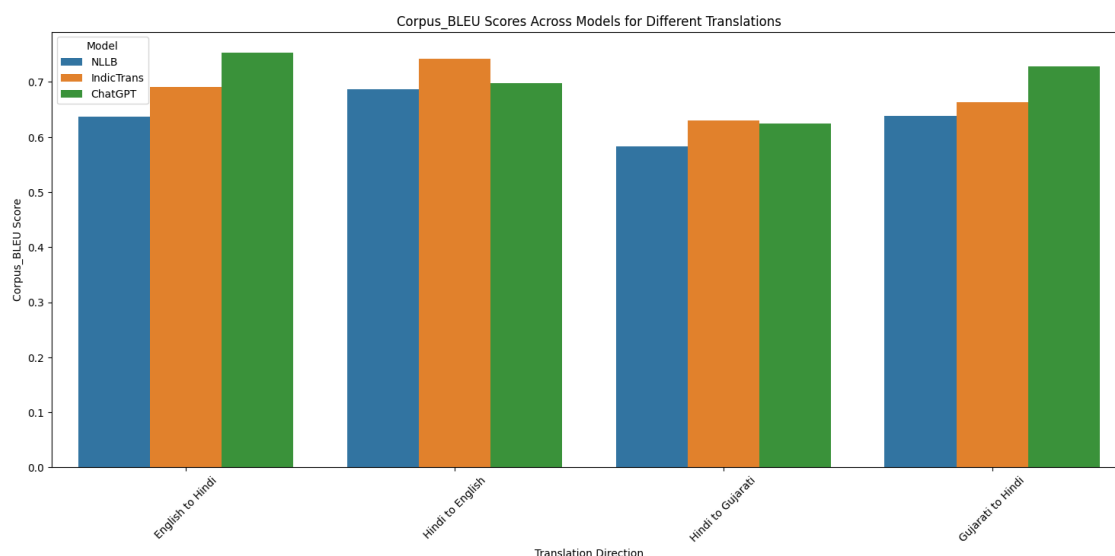
ROUGE_2:-

ROUGE-2 evaluates the quality of text summaries by measuring the overlap of bigrams between the generated summary and reference summaries. It focuses on the co-occurrence of two consecutive words.

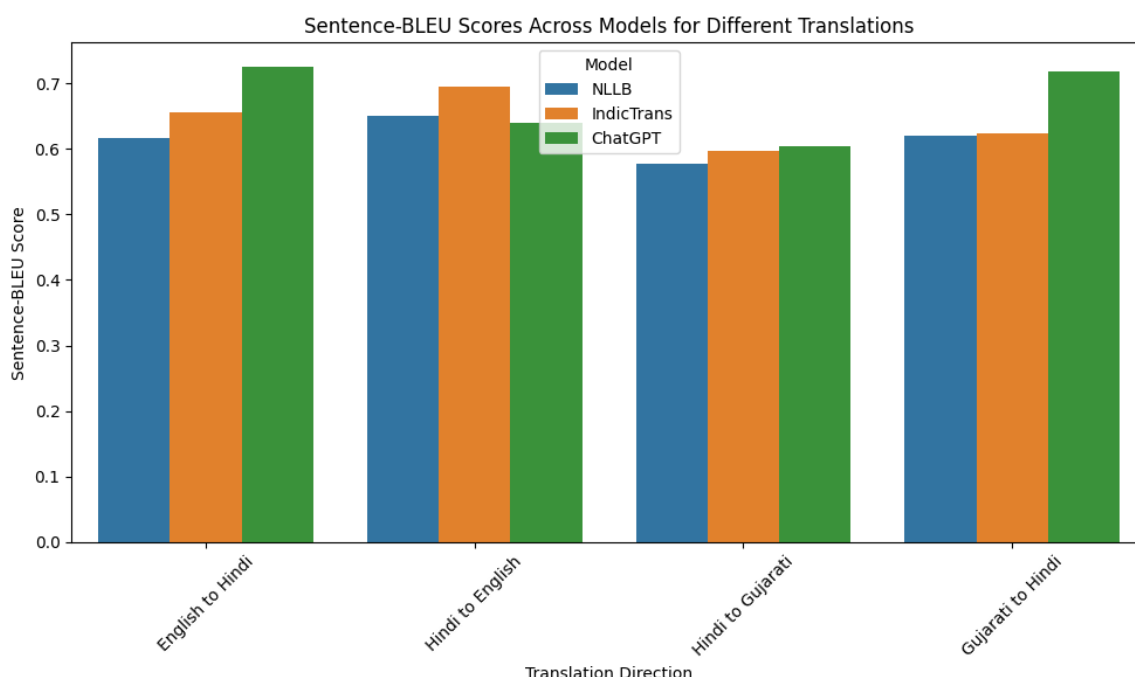
ROUGE_L:-

ROUGE-L uses the longest common subsequence (LCS) between a generated summary and reference texts to evaluate the quality of summaries. It assesses the longest sequence of words that appears in both texts in the same order.

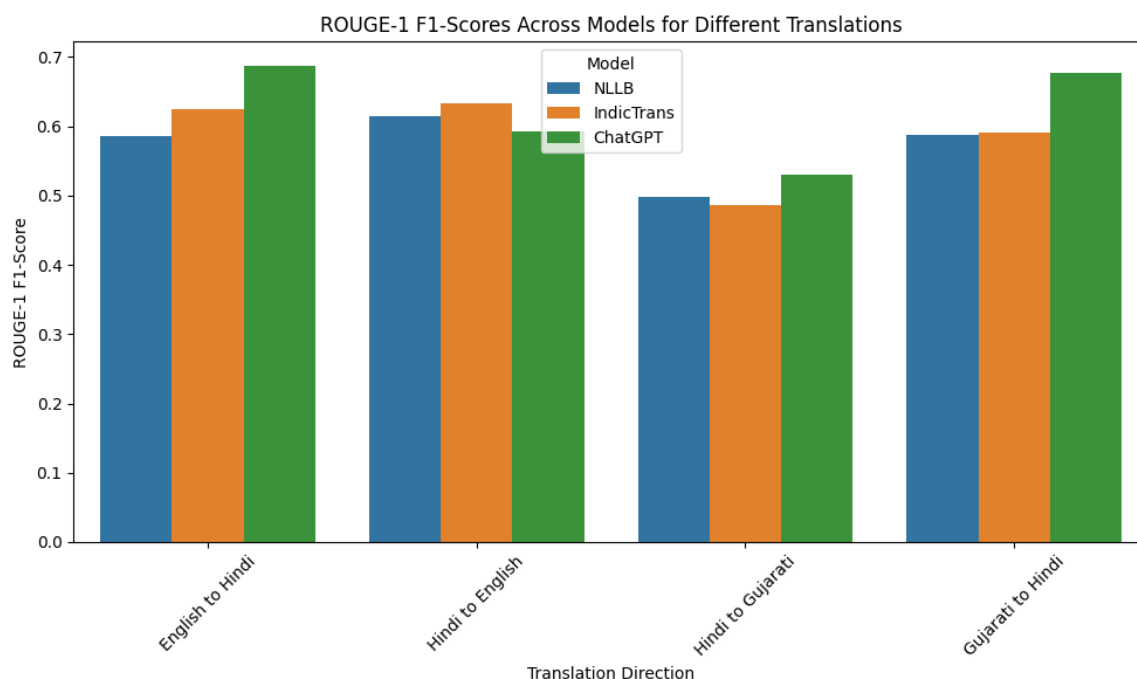
Analyzing each methods one by one:-



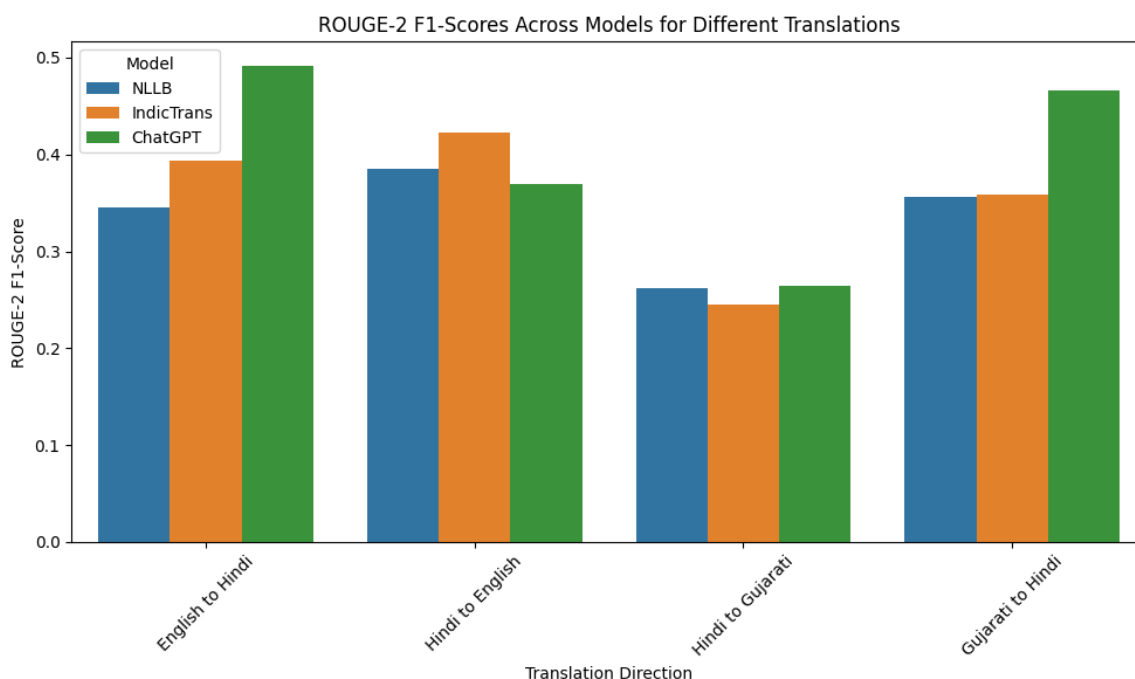
1. **ChatGPT** appears to be the strongest model for the **English to Hindi** translation task, indicating it likely produces translations that most closely match the reference texts for this language pair.
2. For the **Hindi to English** translation, **IndicTrans** shows the highest BLEU score, suggesting that it's the most effective model in translating Hindi sentences into English within the evaluated dataset.
3. When it comes to translating from **Hindi to Gujarati**, **IndicTrans** again leads, indicating a better grasp of translating Hindi context into Gujarati compared to the other models.
4. For **Gujarati to Hindi** translations, **ChatGPT** is again on top, implying that it may handle the translation from Gujarati to Hindi with a higher degree of accuracy and fluency.



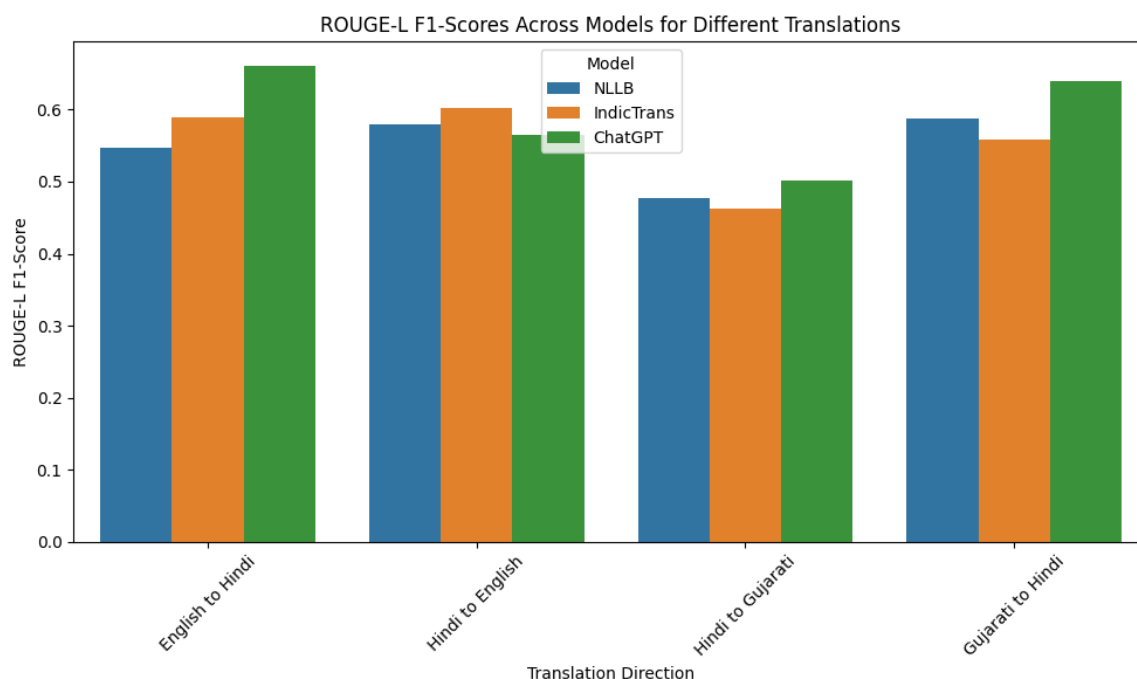
1. **ChatGPT** yields the highest Sentence-BLEU score for **English to Hindi** translations, indicating it may provide the most accurate and fluent translations for this language pair.
2. **IndicTrans** takes the lead in **Hindi to English** translations, suggesting it may better capture the nuances of English when translating from Hindi.
3. **ChatGPT** slightly edges out in **Hindi to Gujarati** translations, indicating it might preserve the meaning between these languages a bit more effectively than others.
4. For **Gujarati to Hindi**, **ChatGPT** shows the highest score, suggesting it has a stronger ability to translate Gujarati context into Hindi accurately.



1. **ChatGPT** excels in the **English to Hindi** translation task, suggesting it captures the overlap with reference summaries well for this language pair.
2. **IndicTrans** shows a slight advantage in **Hindi to English** translations, indicating effective capture of key points in reference texts when translating from Hindi.
3. For **Hindi to Gujarati**, the three models perform comparably, but **ChatGPT** has a marginal lead, implying it may retain essential content slightly better.
4. In **Gujarati to Hindi** translations, **ChatGPT** again scores highest, suggesting its translations are most aligned with the reference summaries for this direction.



1. **ChatGPT** leads in the **English to Hindi** translation, indicating better performance at capturing sequences of words that appear in the reference summaries.
2. For **Hindi to English**, **IndicTrans** slightly outperforms the others, which suggests a stronger ability to retain consecutive word sequences in translation to English.
3. The **Hindi to Gujarati** translation scores are closely clustered, with **ChatGPT** having a slim edge, pointing towards its slightly better translation quality for bigrams.
4. **ChatGPT** dominates the **Gujarati to Hindi** direction, suggesting its superior capacity to translate bigrams or two-word phrases correctly compared to NLLB and IndicTrans.



1. **IndicTrans** shows a competitive edge in **English to Hindi** translation, indicative of its ability to maintain sentence structure and longer sequences of coherent text.
2. In the **Hindi to English** translation task, **ChatGPT** marginally leads, suggesting its capability in preserving the sentence form during translation.
3. For translations from **Hindi to Gujarati**, all models perform comparably, but **ChatGPT** has a slight advantage, which may point to its effective use of the longest common subsequence.
4. **ChatGPT** outperforms the others in translating from **Gujarati to Hindi**, demonstrating its strength in capturing the fluency and flow of the source language into the target language.